

Complex Numbers

Problem 28.1

(1)

$$\begin{aligned}(3 + 2i)(1 - i) \\ 3 - 3i + 2i - 2i^2 \\ 5 - i\end{aligned}$$

(2)

$$\begin{aligned}(1 + i)^4 \\ ((1 + i)^2)^2 \\ (1 + 2i + i^2)^2 \\ (2i)^2 \\ -4\end{aligned}$$

(3)

$$\begin{aligned}i^{100} \\ i^2 = -1 \\ i^4 = 1 \\ (i^4)^{25} = 1\end{aligned}$$

(4)

$$\begin{aligned}(a + bi)(a - bi) \\ a^2 - b^2i^2 - abi + abi \\ a^2 + b^2\end{aligned}$$

Problem 28.2

$$\begin{aligned}(a + bi)^2 = i \\ a^2 - b^2 + 2abi = i \\ (a^2 - b^2) + (2ab)i = 0 + 1i\end{aligned}$$

Equating the real and imaginary parts:

$$\begin{aligned}a^2 - b^2 = 0 \\ 2ab = 1\end{aligned}$$

Solving:

$$\begin{aligned}a = \pm b \\ 2a^2 = 1 \\ a^2 = \frac{1}{2} \\ a = \pm \frac{1}{\sqrt{2}}\end{aligned}$$

$$b = \pm \frac{1}{\sqrt{2}}$$

Problem 28.3

$$\frac{1}{(3-i)} * \frac{(3+i)}{(3+i)}$$

$$= \frac{3+i}{3^2-i^2}$$

$$= \frac{3+i}{10}$$

$$= \frac{3}{10} + \frac{i}{10}$$

Problem 28.5

$$|1| = |1 + 0i| = \sqrt{1^2 + 0^2} = 1$$

$$|i| = |0 + 1i| = \sqrt{0^2 + 1^2} = 1$$

$$|-i| = |0 + (-1)i| = \sqrt{0^2 + (-1)^2} = 1$$

$$|-1| = |-1 + 0i| = \sqrt{(-1)^2 + 0^2} = 1$$

Problem 28.6

$$\text{Given, } |\tau z| = 1$$

$$\tau|z| = 1, \text{ since } \tau \text{ is a real number}$$

$$\tau = \frac{1}{|z|} = \frac{1}{\sqrt{a^2+b^2}}$$

Problem 28.9

$$\begin{bmatrix} 1 \\ 4 \end{bmatrix}$$

Problem 28.10

$$\begin{bmatrix} 0 \\ 2 \end{bmatrix}$$

Problem 28.14

Conjugate or Complex Conjugate